Double Set-Valued Tableaux and K-theory of the Grassmanian

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1 Preliminaries

1.1 Conventions

Fix integers $0 < k < n$ and set $N = n - k$. The *box* is the $k \times N$ rectangle. A partition is a sequence

$$\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$$

of nonnegative integers satisfying

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k.$$

Its Young diagram consists of left-aligned rows of cells, where the i-th row from the top contains λ_i cells. We write

$$\lambda \subseteq \text{box}$$

when this diagram fits inside the $k \times N$ rectangle.

For $\mu \subseteq \text{box}$, the *box complement* is

$$\mu^\vee = (N - \mu_k, N - \mu_{k-1}, \ldots, N - \mu_1),$$

the 180° rotation of the complement of μ in the box. The alphabet is

$$[k] = \{1, 2, \ldots, k\}.$$

1.2 Semistandard Young tableaux

Definition 1.1 (Semistandard Young tableau)**.** *Let λ be a possibly skew shape. A* semistandard Young tableau *(SSYT) of shape λ is a filling of its cells by entries from $[k]$ such that the entries weakly increase from left to right along each row and strictly increase from top to bottom down each column.*

If the letter i occurs c_i times, then

$$\text{content}(T) = (c_1, \ldots, c_k)$$

is the content of T, and its weight is

$$\mathbf{x}^{\text{content}(T)} = x_1^{c_1} \cdots x_k^{c_k}.$$

Example 1.2. *The following is an SSYT of shape* $(4, 3, 2)$ *with* $k = 4$*:*

1	1	2	2
2	3	3	
3	4		

Its content is

$$\mathrm{content}(T) = (2, 3, 3, 1),$$

and hence its weight is

$$\mathbf{x}^{\mathrm{content}(T)} = x_1^2 x_2^3 x_3^3 x_4.$$

1.3 Set-valued tableaux

Definition 1.3 (Set-valued tableau)**.** *Let* λ *be a possibly skew shape. A* set-valued semistandard Young tableau *(SVT) of shape* λ *is a filling that assigns to each cell a nonempty subset of* $\{1, \ldots, k\}$ *such that, writing* A *for the set in a cell,*

(S1) (rows)

$$\max A \leq \min B$$

whenever B *is the cell immediately to the right of* A*;*

(S2) (columns)

$$\max A < \min C$$

whenever C *is the cell immediately below* A*.*

Write $|T|$ *for the total number of letters, with multiplicity across cells, and write*

$$\mathrm{content}(T) = (c_1, \ldots, c_k),$$

where c_i *is the number of occurrences of* i*. Define*

$$\mathrm{exc}(T) = |T| - |\lambda|$$

to be the excess*, namely the number of extra letters beyond one per cell. When every cell is a singleton,* T *is an ordinary semistandard Young tableau.*

Remark. We draw a cell holding the set

$$\{a < b < \cdots\}$$

by writing its elements in increasing order with no separator. Thus a cell containing 12 denotes the set $\{1, 2\}$.

Example 1.4. *A set-valued tableau of shape* $(4, 3, 2)$ *with* $k = 4$ *is*

1	1	2	23
2	3	3	
34	4		

Two cells are genuinely set-valued, namely the cells containing $\{2,3\}$ *and* $\{3,4\}$*. Therefore*

$$|T| = 11, \qquad |\lambda| = 9, \qquad \mathrm{exc}(T) = 2,$$

and its content is

$$\mathrm{content}(T) = (2,3,4,2).$$

A *reverse set-valued tableau* of shape λ is the value complement

$$v \longmapsto k+1-v$$

of a set-valued tableau. Equivalently, along each row the entries weakly decrease from left to right and down each column they strictly decrease.

In the set-valued notation, this means

$$\min A \geq \max B$$

whenever B is immediately to the right of A, and

$$\min A > \max C$$

whenever C is immediately below A. Its excess and content are defined in the same way as for an ordinary set-valued tableau.

2 DSVTs and MSSYTs

2.1 Doubled set-valued tableaux

Definition 2.1 (Doubled set-valued tableau)**.** *Fix* $\lambda, \mu \subseteq \mathrm{box}$ *and suppose that*

$$\lambda \subseteq \mu^\vee.$$

A doubled set-valued tableau *(DSVT) for the box* $k \times N$ *is a filling of the box in which*

- *the* λ*-region at the top-left carries a reverse set-valued tableau of shape* λ*; and*
- *the* μ*-region at the bottom-right carries a* $180°$*-rotated set-valued tableau of shape* μ*.*

The two regions are compatible along the cylinder. More precisely, if one row meets both regions, let A *be the rightmost cell of the* μ*-region and let* B *be the leftmost cell of the* λ*-region. The horizontal seam condition is*

$$\min A \geq \max B.$$

If one column meets both regions, let C *be the lowest cell of the* μ*-region and let* D *be the highest cell of the* λ*-region. The vertical seam condition is*

$$\min C > \max D.$$

The cells not occupied by the two regions form the skew shape

$$\theta = \mu^\vee/\lambda.$$

The excess of a DSVT is the total number of set-valued surplus letters, and its content is the total content:

$$\operatorname{exc}(D) = \sum_{A \in D} \bigl(|A| - 1\bigr).$$

Example 2.2. *Take*

$$k = 4, \qquad N = 6, \qquad \lambda = (3, 1, 1), \qquad \mu = (4, 3, 1).$$

Then

$$\mu^{\vee} = (6, 5, 3, 2) \supseteq \lambda.$$

A doubled set-valued tableau is

4	3	2			
3					34
12			234	2	2
		34	1	1	1

For example, the horizontal seam condition in the third row is

$$\min\{2\} \geq \max\{1, 2\},$$

while the vertical seam condition in the third column is

$$\min\{3, 4\} > \max\{2\}.$$

The excess of this DSVT is

$$\operatorname{exc}(D) = 5.$$

2.2 Marked semistandard Young tableaux

Marked tableaux are filled with the numbers $1, \ldots, k$, each entry being either plain or marked. We write a marked entry of value v as the primed letter v'.

Definition 2.3 (Marked semistandard Young tableau)**.** *Let θ be a skew shape. A* marked semistandard Young tableau *(MSSYT) of shape θ is a filling of its cells by plain or primed letters from*

$$1' < 1 < 2' < 2 < \cdots < k' < k.$$

For $x \in \{v, v'\}$, write

$$|x| = v$$

for its numerical value. The filling is required to satisfy the following conditions.

(G1) (columns) *the numbers strictly increase from top to bottom (marks are invisible to this rule), so each value occurs at most once in a column;*

(G2) (rows) *the numbers weakly increase from left to right, and a marked entry is strictly larger than its left neighbour (so a given value is marked at most once in a row);*

(G3) (flag) *a marked entry of value v lying in the d-th cell of its column of θ (counted from the top) satisfies*

$$v > d.$$

The weight records the plain content, and the number of primed entries is the degree:

$$\mathbf{x}^{\mathrm{content}(G)} = \prod_{\substack{x \in G \\ x \text{ plain}}} x_{|x|}, \qquad \mathrm{exc}(G) = \#\{x \in G : x \text{ is primed}\}.$$

Equivalently, forgetting the marks leaves an ordinary column-strict, row-weak increasing filling of θ, namely a skew SSYT, and a mark records a value justified by a missing value in the column completion. In particular, the value 1 can never be marked, and one cannot mark v in a column in which every value smaller than v already occurs plain. Thus marked SSYTs form a proper subset of skew SSYTs with marks placed on an arbitrary subset of cells. We write a marked value v as the primed letter v' and use this primed notation throughout.

Example 2.4. *Let*

$$k = 4, \qquad N = 6, \qquad \theta = \mu^\vee/\lambda = (6,5,3,2)/(3,1,1),$$

where

$$\lambda = (3,1,1), \qquad \mu = (4,3,1).$$

The following tableau is an MSSYT of shape θ:

			$3'$	3	$4'$
	1	1	$4'$	4	
	2	$4'$			
$2'$	4				

Its plain content and excess are

$$\mathbf{x}^{\mathrm{content}(G)} = x_1^2 x_2 x_3 x_4^2, \qquad \mathrm{exc}(G) = 5.$$

Therefore its contribution to the marked Schur function is

$$\beta^5 x_1^2 x_2 x_3 x_4^2.$$

2.3 The column-completion bijection

A doubled set-valued tableau and its MSSYT live on the same $k \times N$ grid. The blue cells are the doubled tableau: a reverse set-valued tableau of λ at the top-left and a rotated set-valued tableau of μ at the bottom-right. The red cells form the MSSYT on

$$\mu^\vee/\lambda.$$

The MSSYT is obtained by column completion. In each DSVT cell, retain its minimum. The other entries of the cell are its set-valued surplus entries. In each column, place the values missing from the set of minima into the complementary cells. A missing value is primed precisely when it occurs as a nonminimal entry in the original DSVT column.

In each completed column, the DSVT minima and the complementary entries together contain every value in

$$[k] = \{1, \dots, k\}$$

exactly once. This is the content-complementing property of the bijection.

Example 2.5 (Column completion in one column)**.** *For $k = 5$, consider a DSVT column with set-valued cells*

$$B_1 = \{1, 3\}, \qquad B_2 = \{4, 5\}.$$

Its minima are 1 and 4, its nonminimal entries are 3 and 5, and the missing values are

$$[5] \setminus \{1, 4\} = \{2, 3, 5\}.$$

Therefore column completion produces the MSSYT column

$$\begin{array}{|c|}\hline 2 \\ \hline 3' \\ \hline 5' \\ \hline \end{array}.$$

The DSVT column has excess two, and the completed column has exactly two primed entries.

Example 2.6 (Complete column completion)**.** *The DSVT in Example 2.2 and its corresponding MSSYT can be displayed on the same 4×6 grid:*

4	3	2	$3'$	3	$4'$
3	1	1	$4'$	4	34
12	2	$4'$	234	2	2
$2'$	4	34	1	1	1

Reading only the red cells gives

			$3'$	3	$4'$
	1	1	$4'$	4	
	2	$4'$			
$2'$	4				

which is exactly the MSSYT in Example 2.4.

Theorem 2.7 (DSVT–MSSYT bijection)**.** *There is a bijection*

$$\varphi : \mathrm{DSVT}(\lambda, \mu) \longrightarrow \mathrm{MSSYT}(\mu^\vee/\lambda).$$

If $G = \varphi(D)$, *then*

$$\operatorname{exc}(G) = \operatorname{exc}(D)$$

and

$$\operatorname{content}(G) = (N, \dots, N) - \operatorname{content}(D).$$

Here $\operatorname{content}(G)$ *counts only the plain entries of* G*.*

Proof. Let D be a DSVT. In one of its columns, write the set-valued cells as

$$B_1, \dots, B_t$$

and put

$$m_i = \min B_i.$$

Replace each B_i by m_i. Doing this in every column gives an ordinary doubled tableau $D^{\min}$.

Apply the ordinary sandwich bijection to $D^{\min}$. In each column, the values

$$[k] \setminus \{m_1, \dots, m_t\}$$

are placed, in increasing order, into the cells of $\mu^\vee/\lambda$. This gives an ordinary skew SSYT U.

If a completed cell contains v, mark it exactly when v occurs as a nonminimal entry in the original DSVT column; that is, when

$$v \in B_i \setminus \{m_i\}$$

for some i. Write the marked value as v', and denote the resulting filling by $\varphi(D)$.

This rule is well defined because the column condition of a DSVT prevents the same value from occurring in two different cells of one column.

After the marks are forgotten, $\varphi(D)$ is the skew SSYT U. Thus (G1) and the weakly increasing part of (G2) follow from the ordinary sandwich bijection.

Suppose that a marked entry v' has a left neighbour of numerical value v. Let B be the DSVT cell corresponding to the left completed entry, and let C be the DSVT cell containing v as a nonminimal entry in the right column. Since the two completed entries have the same value and lie in the same skew row, the completion-rank property places B and C in the same row of the doubled tableau. Moreover,

$$\min B < v \qquad \text{and} \qquad v \leq \max C.$$

The DSVT row condition gives

$$\min B \geq \max C,$$

and hence

$$v > \min B \geq \max C \geq v,$$

a contradiction. Therefore a marked entry is strictly larger than its left neighbour, proving (G2).

Now let v' be the d-th cell of its skew column. The $d-1$ entries above it have distinct numerical values smaller than v. Since v is a nonminimal entry of some DSVT cell B, the value $\min B$ is another value smaller than v, and it is not one of these completed entries. There are therefore at least d distinct values smaller than v. Since $[k]$ contains exactly $v-1$ values smaller than v, we have

$$d \leq v-1,$$

so

$$v > d.$$

This proves (G3), and hence $\varphi(D)$ is an MSSYT.

The construction also gives the weight relation. In each column, the DSVT entries together with the plain entries of $\varphi(D)$ contain every value in $[k]$ exactly once. Indeed, the nonminimal DSVT entries are exactly the completed entries that are marked and therefore do not contribute to the plain content. Since the box has N columns,

$$\mathrm{content}(D) + \mathrm{content}(\varphi(D)) = (N, \ldots, N).$$

Moreover, every nonminimal DSVT entry produces one marked entry, so

$$\mathrm{exc}(\varphi(D)) = \mathrm{exc}(D).$$

We now construct the inverse map. Let G be an MSSYT. Forget the marks and let U be the resulting skew SSYT. The inverse ordinary sandwich bijection recovers a unique ordinary doubled tableau D_0. Regard each cell of D_0 as a singleton set.

Let v' be a marked entry of G. In the corresponding column of D_0, insert v into the unique cell B such that

$$\min B < v < \min C,$$

where C is the cell immediately above B. If B is the highest cell of that column, we use the convention

$$\min C = k+1.$$

To see that this cell exists, suppose that v' is the d-th cell of its skew column. There are $d-1$ completed entries above it, all smaller than v. Condition (G3) gives

$$v > d,$$

so at least one value in $\{1, \ldots, v-1\}$ is not among these completed entries. It therefore occurs in the ordinary doubled tableau. The completion-rank property then determines the unique cell B for which

$$\min B < v < \min C.$$

Inserting v into B preserves the column condition.

It remains to check the row condition. Let A and B be adjacent cells, with A on the left. Suppose that an inserted value $v \in B$ violates the row condition, so that

$$\min A < v.$$

Let C be the cell above B and let E be the cell above A. The insertion rule and the ordinary row inequalities give

$$\min B \leq \min A < v < \min C \leq \min E.$$

Consequently, the completed left column contains an entry of value v in the same skew row as the marked entry v' in the right column. This contradicts (G2). Hence the reconstructed filling is a DSVT.

The inverse construction retains the same cell minima and inserts each marked value into the cell from which it arose. Therefore

$$\psi(\varphi(D)) = D.$$

Conversely, applying φ after the inverse construction marks exactly the values that were inserted, so

$$\varphi(\psi(G)) = G.$$

Thus φ and ψ are mutually inverse. $\square$

Corollary 2.8 (Weight complementation)**.** *Let*

$$\mathfrak{g}_{\mu^\vee/\lambda}(\mathbf{x};\beta) = \sum_{G \in \mathrm{MSSYT}(\mu^\vee/\lambda)} \beta^{\mathrm{exc}(G)} \mathbf{x}^{\mathrm{content}(G)}.$$

Then

$$\mathfrak{g}_{\mu^\vee/\lambda}(\mathbf{x};\beta) = (x_1 \cdots x_k)^N \sum_{D \in \mathrm{DSVT}(\lambda,\mu)} \beta^{\mathrm{exc}(D)} \mathbf{x}^{-\,\mathrm{content}(D)}.$$

Proof. For $G = \varphi(D)$, Theorem 2.7 gives

$$\mathrm{exc}(G) = \mathrm{exc}(D)$$

and

$$\mathbf{x}^{\mathrm{content}(G)} = (x_1 \cdots x_k)^N \mathbf{x}^{-\,\mathrm{content}(D)}.$$

The result follows by summing over all DSVTs. $\square$

Define

$$\mathcal{D}_N(f)(x_1,\ldots,x_k) = (x_1 \cdots x_k)^N f(x_1^{-1},\ldots,x_k^{-1}).$$

The preceding corollary becomes

$$\mathfrak{g}_{\mu^\vee/\lambda} = \mathcal{D}_N\left(\sum_D \beta^{\mathrm{exc}(D)} \mathbf{x}^{\mathrm{content}(D)}\right).$$

Corollary 2.9 (Schur positivity)**.** *The marked Schur function*

$$\mathfrak{g}_{\mu^\vee/\lambda}(\mathbf{x};\beta)$$

is Schur-positive.

Proof. Let

$$T_{\lambda,\mu}(\mathbf{x};\beta) = \sum_D \beta^{\mathrm{exc}(D)} \mathbf{x}^{\mathrm{content}(D)}$$

be the DSVT generating function. By weight complementation,

$$\mathfrak{g}_{\mu^\vee/\lambda} = \mathcal{D}_N(T_{\lambda,\mu}),$$

where

$$\mathcal{D}_N(f)(x_1, \ldots, x_k) = (x_1 \cdots x_k)^N f(x_1^{-1}, \ldots, x_k^{-1}).$$

Write the Schur expansion of $T_{\lambda,\mu}$ as

$$T_{\lambda,\mu} = \sum_{\nu \subseteq \mathrm{box}} a_\nu(\beta) s_\nu, \qquad a_\nu(\beta) \in \mathbb{Z}_{\geq 0}[\beta].$$

For every $\nu \subseteq$ box,

$$\mathcal{D}_N(s_\nu) = s_{\nu^\vee}.$$

It follows that

$$\mathfrak{g}_{\mu^\vee/\lambda} = \sum_{\nu \subseteq \mathrm{box}} a_\nu(\beta) s_{\nu^\vee},$$

so $\mathfrak{g}_{\mu^\vee/\lambda}$ is Schur-positive. ☐

Corollary 2.10 (Marked Schur expansion)**.** *Write the box-truncated product as*

$$G_\lambda G_\mu\big|_{\mathrm{box}} = \sum_{\nu \subseteq \mathrm{box}} c^\nu_{\lambda\mu}(\beta) G_\nu.$$

Then

$$\mathfrak{g}_{\mu^\vee/\lambda} = \sum_{\nu \subseteq \mathrm{box}} c^\nu_{\lambda\mu}(\beta) \mathfrak{g}_{\nu^\vee}.$$

Thus the coefficients in the expansion of a skew marked Schur function into straight marked Schur functions are the K-theoretic Littlewood–Richardson coefficients.

Proof. The DSVT product formula and weight complementation give

$$\mathfrak{g}_{\mu^\vee/\lambda} = \mathcal{D}_N \left(G_\lambda G_\mu\big|_{\mathrm{box}} \right).$$

For a straight shape ν, column completion gives

$$\mathcal{D}_N(G_\nu) = \mathfrak{g}_{\nu^\vee}.$$

Therefore

$$\begin{aligned} \mathfrak{g}_{\mu^\vee/\lambda} &= \mathcal{D}_N \left(\sum_{\nu \subseteq \mathrm{box}} c^\nu_{\lambda\mu}(\beta) G_\nu \right) \\ &= \sum_{\nu \subseteq \mathrm{box}} c^\nu_{\lambda\mu}(\beta) \mathcal{D}_N(G_\nu) \\ &= \sum_{\nu \subseteq \mathrm{box}} c^\nu_{\lambda\mu}(\beta) \mathfrak{g}_{\nu^\vee}. \end{aligned}$$

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3 Marked Schur Functions and Bender–Knuth Attempts

The preceding section established an explicit bijection between DSVTs and MSSYTs of the corresponding complementary skew shape. The bijection does more than identify two classes of tableaux. It translates the set-valued information in a DSVT into ordinary entries together with prime data in an MSSYT. The weight relation established in the preceding section consequently converts the relevant DSVT generating function into a generating function over MSSYTs. This leads naturally to the following definition.

Definition 3.1 (Marked Schur function)**.** *For partitions λ and μ, let*

$$\theta = \mu^\vee/\lambda.$$

The marked Schur function associated with θ is

$$\mathfrak{g}_\theta(x;\beta) = \sum_{G\in \mathrm{MSSYT}(\theta)} \beta^{\mathrm{exc}(G)} x^{\mathrm{content}(G)},$$

where G ranges over all MSSYTs of shape θ. Here $\mathrm{content}(G)$ *records the multiplicities of the plain entries, while* $\mathrm{exc}(G)$ *is the number of primed entries.*

The next question is whether this generating function is symmetric in the variables $x_1, x_2, \ldots$. In the ordinary semistandard setting, symmetry is proved combinatorially by the Bender–Knuth involutions, which exchange the multiplicities of two adjacent letters i and $i+1$. This suggested searching for an analogous operation on MSSYTs.

The presence of primed entries creates two additional requirements. Such an operation must preserve the number of primes, since this is the exponent of β, and it must preserve the marked tableau conditions (G1)–(G3). In particular, moving a prime can affect the strict-neighbour condition in (G2) and the depth restriction in (G3).

Motivated by the ordinary Bender–Knuth construction, we therefore sought, for each i, a mark-preserving involution

$$\widetilde{BK}_i : \mathrm{MSSYT}(\theta) \longrightarrow \mathrm{MSSYT}(\theta)$$

satisfying

$$\mathrm{wt}_i(T) = \mathrm{wt}_{i+1}\big(\widetilde{BK}_i(T)\big), \qquad \mathrm{wt}_{i+1}(T) = \mathrm{wt}_i\big(\widetilde{BK}_i(T)\big).$$

All other plain multiplicities and the total number of primed entries should remain unchanged. Such an involution would pair the terms whose weights differ by exchanging x_i and x_{i+1} and would therefore give a direct combinatorial proof of symmetry.

Two successive constructions were developed. The first used six local vertical configurations followed by an ordinary Bender–Knuth exchange and row reordering. The second refined this rule by recording the relative positions of the first relevant entries and processing eight types of local blocks from bottom to top.

3.1 The Six-Configuration Construction

The first construction examines the configurations in which an entry i or i' lies immediately above an entry $i+1$ or $(i+1)'$. Six local configurations were distinguished.

Case 1. The plain vertical pair

$$\begin{array}{|c|}\hline i \\ \hline i+1 \\ \hline\end{array}$$

is kept fixed:

$$\begin{array}{|c|}\hline i \\ \hline i+1 \\ \hline\end{array} \longmapsto \begin{array}{|c|}\hline i \\ \hline i+1 \\ \hline\end{array}.$$

In the original formulation, this case was used when $(i+1)'$ did not occur in the relevant part of the lower row, or when the upper i occupied depth d and $i \leq d$. The latter condition was intended to prevent the rule from turning this i into an illegal primed entry.

Case 2. Suppose that the upper distinguished entry is a plain i, while the lower block begins with $(i+1)'$ and ends with a plain $i+1$:

$$\begin{array}{ccc}\cline{3-3} & & \multicolumn{1}{|c|}{i} \\ \hline \multicolumn{1}{|c|}{(i+1)'} & \multicolumn{1}{c|}{\cdots} & \multicolumn{1}{c|}{i+1} \\ \hline\end{array}.$$

When the upper i occupies depth d and $i > d$, the proposed transformation is

$$\begin{array}{ccc}\cline{3-3} & & \multicolumn{1}{|c|}{i} \\ \hline \multicolumn{1}{|c|}{(i+1)'} & \multicolumn{1}{c|}{\cdots} & \multicolumn{1}{c|}{i+1} \\ \hline\end{array} \longmapsto \begin{array}{ccc}\cline{3-3} & & \multicolumn{1}{|c|}{i'} \\ \hline \multicolumn{1}{|c|}{i+1} & \multicolumn{1}{c|}{\cdots} & \multicolumn{1}{c|}{i+1} \\ \hline\end{array}.$$

Thus the prime is moved from the lower distinguished entry to the upper distinguished entry.

Case 3. The vertical pair in which both distinguished entries are primed is kept fixed:

$$\begin{array}{|c|}\hline i' \\ \hline (i+1)' \\ \hline\end{array} \longmapsto \begin{array}{|c|}\hline i' \\ \hline (i+1)' \\ \hline\end{array}.$$

Case 4. When the upper entry is a plain i and the lower entry is $(i+1)'$, the prime is transferred upward:

$$\begin{array}{|c|}\hline i \\ \hline (i+1)' \\ \hline\end{array} \longmapsto \begin{array}{|c|}\hline i' \\ \hline i+1 \\ \hline\end{array}.$$

Case 5. The reverse prime transfer is

$$\begin{array}{|c|}\hline i' \\ \hline i+1 \\ \hline\end{array} \longmapsto \begin{array}{|c|}\hline i \\ \hline (i+1)' \\ \hline\end{array}.$$

The original rule used this transformation when the relevant lower row did not contain another $(i+1)'$.

Case 6. The final configuration has the form

$$\begin{array}{ccc} \cline{3-3} & & \multicolumn{1}{|c|}{i'} \\ \hline \multicolumn{1}{|c|}{(i+1)'} & \multicolumn{1}{c|}{\cdots} & \multicolumn{1}{c|}{i+1} \\ \hline \end{array}.$$

In this case, the entries i' and $(i+1)'$ in the two relevant rows are kept primed. The plain entries i and $i+1$ are then transformed according to the ordinary Bender–Knuth multiplicity exchange.

After one of the first five local transformations has been applied, the selected vertical column is kept fixed. In all remaining cells, the primed entries i' and $(i+1)'$ remain unchanged, while the plain entries are transformed according to the ordinary Bender–Knuth rule.

If several consecutive columns contain an entry i or i' above an entry $i+1$ or $(i+1)'$, the proposed rule keeps all but the leftmost distinguished column fixed and applies the prime change only in the leftmost column. Finally, the entries in every affected row are rearranged so that the weak row order is restored.

Attempt 1 (Attempted involution check)**.** *We will prove that applying $\widetilde{BK}_i$ twice recovers the original tableau.*

For Cases 1 and 3, applying the transformation twice returns to the original tableau.

For Case 2, after the transformation, the tableau falls into Case 5. We then remove the prime from a' and add a prime to $a+1$, which recovers the original tableau.

For Case 4, after the transformation, the second row contains no entry $(a+1)'$. Therefore, when applying the local transformation again, the tableau falls into Case 5. We remove the prime from a' and add a prime to $a+1$, which recovers the original tableau.

For Case 5, there are two possible subcases.

1. *Suppose that the second row of the original tableau does not contain any entry a. After the transformation, the tableau becomes*

$$\begin{array}{ccc} \cline{3-3} & & \multicolumn{1}{|c|}{a} \\ \hline \multicolumn{1}{|c|}{\cdots} & \multicolumn{1}{c|}{a} & \multicolumn{1}{c|}{(a+1)'} \\ \hline \end{array}.$$

We are now in Case 4. We put a prime on a and remove the prime from $a+1$. After applying the usual Bender–Knuth involution, we recover the original tableau:

$$\begin{array}{ccc} \cline{3-3} & & \multicolumn{1}{|c|}{a'} \\ \hline \multicolumn{1}{|c|}{\cdots} & \multicolumn{1}{c|}{a+1} & \multicolumn{1}{c|}{a+1} \\ \hline \end{array}.$$

2. Suppose that the second row of the original tableau contains an entry a. After the transformation, the tableau becomes

$$\begin{array}{cccc} & & & \boxed{a} \\ \boxed{\cdots} & \boxed{(a+1)'} & \boxed{\cdots} & \boxed{a+1} \end{array}.$$

We are now in Case 2. Applying the transformation a second time recovers the original tableau.

For Case 6, there are also two possible subcases.

1. Suppose that the second row of the original tableau does not contain any entry a. After the transformation, the tableau becomes

$$\begin{array}{cc} & \boxed{a'} \\ \boxed{a} & \boxed{(a+1)'} \end{array}.$$

In this situation, we are in Case 3. Applying the transformation again, we keep a' and $(a+1)'$ fixed, change a into $a+1$, and reorder the entries in the row. This recovers the original tableau.

2. Suppose that the second row of the original tableau contains an entry a. After the transformation, the tableau becomes

$$\begin{array}{ccc} & & \boxed{a'} \\ \boxed{(a+1)'} & \boxed{\cdots} & \boxed{a+1} \end{array}.$$

In this situation, we are again in Case 6. Keeping a' and $(a+1)'$ fixed, we exchange the numbers of occurrences of a and $a+1$, and then reorder the entries.

After applying the transformation twice, the first row still contains a', the second row still contains $(a+1)'$, and the numbers of occurrences of a and $a+1$ return to their original values. Hence the original tableau is recovered.

Therefore, the local case analysis suggests that applying $\widetilde{BK}_i$ twice recovers the original tableau.

The case-by-case calculation verifies some isolated local configurations, but does not prove that the rule is a well-defined self-map of all MSSYTs. The choice of a consecutive vertical block can change after the prescribed row reordering, and the check does not control the global column inequalities, the strict-neighbour part of (G2), or the flag (G3). Thus applying the rule twice need not encounter the inverse block decomposition. We therefore retain the six cases as a record of the first construction, not as a theorem.

Example 3.2.

$$\begin{array}{ccccccccc} & & & & & & \boxed{2} & \boxed{2} & \boxed{2} \\ & & & & \boxed{2} & \boxed{3} & \boxed{3'} & & \\ \boxed{2'} & \boxed{2} & \boxed{2} & \boxed{3'} & \boxed{3} & & & & \end{array} \xrightarrow{local} \begin{array}{ccccccccc} & & & & & & \boxed{2'} & \boxed{2} & \boxed{2} \\ & & & & \boxed{2'} & \boxed{3} & \boxed{3} & & \\ \boxed{2'} & \boxed{2} & \boxed{2} & \boxed{3} & \boxed{3} & & & & \end{array}$$

$$\xrightarrow{\mathrm{BK}_2} \begin{array}{ccccccccc} & & & & & & \boxed{2'} & \boxed{3} & \boxed{3} \\ & & & & \boxed{2'} & \boxed{2} & \boxed{3} & & \\ \boxed{2'} & \boxed{2} & \boxed{3} & \boxed{3} & \boxed{3} & & & & \end{array}$$

3.2 The Bottom-Up Eight-Case Construction

The preceding obstruction suggested that the relative positions of the distinguished entries had to be included in the local data. We therefore introduced a second construction that processes the tableau from bottom to top. At each stage, in the lowest untreated pair of rows, we locate the first occurrence of i or i' in the upper row and the first occurrence of $i+1$ or $(i+1)'$ in the lower row. The resulting block is of Type A when these two distinguished entries lie in the same column, and of Type B when the distinguished lower entry lies strictly to the left of the distinguished upper entry. Their priming states then divide Types A and B into eight cases.

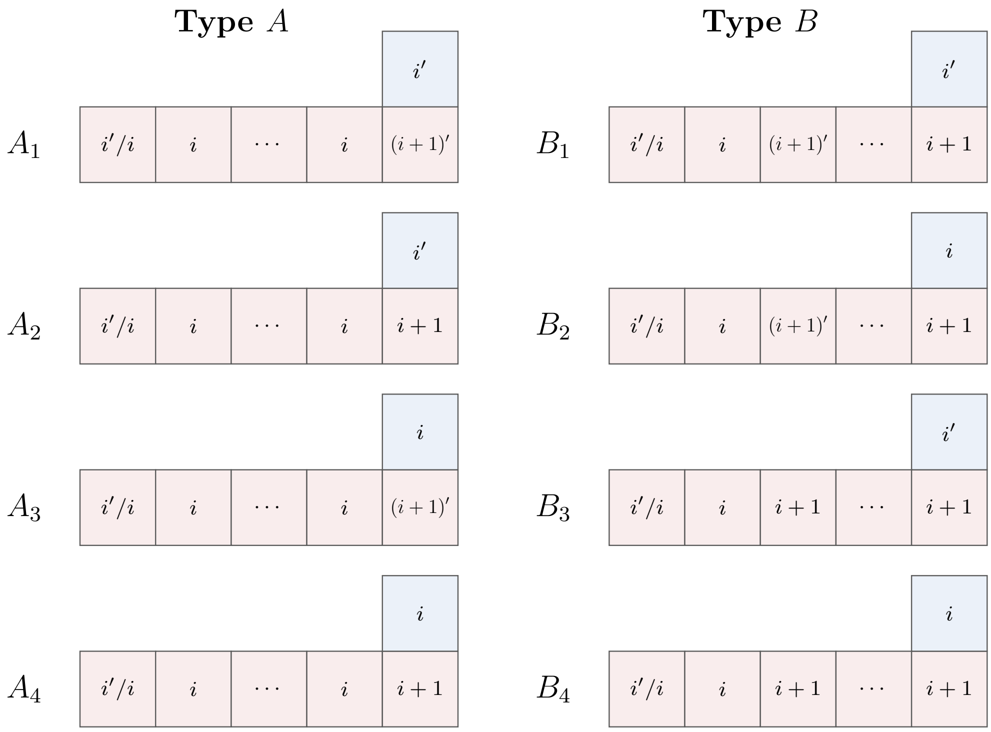

Figure 1: The eight local cases. Here i'/i means that the first lower occurrence of i may be marked or plain. The dots represent an arbitrary admissible horizontal run and impose no additional hypothesis.

The four priming patterns are

priming pattern	Type A	Type B
both distinguished entries primed	A_1	B_1
only the upper entry primed	A_2	B_3
only the lower entry primed	A_3	B_2
both distinguished entries plain	A_4	B_4

The common rule for A_1 and B_1. The relevant lower block has a unique expression

$$(i')^{\varepsilon} i^{a} (i+1)' (i+1)^{b}, \qquad \varepsilon \in \{0,1\}, \qquad a, b \geq 0.$$

The proposed transformation is

$$(i')^{\varepsilon} i^{a} (i+1)' (i+1)^{b} \longmapsto (i')^{\varepsilon} i^{b} (i+1)' (i+1)^{a}.$$

Thus the multiplicities of the plain entries i and $i+1$ are exchanged. The marked entries retain their marks, and $(i+1)'$ becomes the first entry of numerical value $i+1$ in the reordered lower block. The condition $b=0$ corresponds to Type A_1, while $b>0$ corresponds to Type B_1. Hence the same formula is intended to cover both directions even when the geometric type changes.

The direct cases A_4, B_2, B_4. In these three cases, the terminal vertical column is kept fixed. The transformation acts on the part of the lower row lying strictly to the left of that column. The multiplicities of all plain i's and plain $i+1$'s in this part are exchanged, after which the affected interval is refilled in weakly increasing order. In Case B_2, the marked entry $(i+1)'$ is also kept fixed.

Case A_2. The terminal column is

$$\begin{array}{|c|}\hline i' \\ \hline i+1 \\ \hline\end{array}.$$

When the lower block contains a plain i, let a and b be the total numbers of plain i's and plain $i+1$'s in the block, including the terminal $i+1$. Retain one plain i together with the terminal plain $i+1$. Exchange the remaining multiplicities

$$(a-1, b-1) \longmapsto (b-1, a-1),$$

and then restore weak row order. The total multiplicities therefore change from (a,b) to (b,a).

When the lower block contains no plain i, the proposed rule pairs this configuration with the boundary subcase of A_3. A lower marked i', when present, is retained, and the terminal column is changed by

$$\begin{array}{|c|}\hline i' \\ \hline i+1 \\ \hline\end{array} \longleftrightarrow \begin{array}{|c|}\hline i \\ \hline (i+1)' \\ \hline\end{array}.$$

Case A_3. The terminal column is

$$\begin{array}{|c|}\hline i \\ \hline (i+1)' \\ \hline\end{array}.$$

Suppose that the current lower block contains $c \geq 0$ plain i's before this terminal column. Replace the terminal column by

$$\begin{array}{|c|}\hline i' \\ \hline i+1 \\ \hline\end{array},$$

change the preceding c plain i's into c plain $i+1$'s, and restore weak row order.

When $c=0$, the output is the no-plain-i subcase of A_2. When $c>0$, the output is the no-plain-i subcase of B_3.

Case B_3. When the lower block contains a plain i, the rule is the same retained balanced-pair construction used in A_2. The terminal column

$$\begin{array}{|c|}\hline i' \\ \hline i{+}1 \\ \hline\end{array}$$

and one lower plain i are retained, while the remaining plain multiplicities are exchanged.

When the lower block contains no plain i, this case is intended to be the inverse of A_3 with $c > 0$. Change the terminal column back to

$$\begin{array}{|c|}\hline i \\ \hline (i{+}1)' \\ \hline\end{array},$$

and change all other plain $i + 1$'s in the current block into plain i's.

After a local block has been processed, every following consecutive vertical column of the form

$$\begin{array}{|c|}\hline i \\ \hline i{+}1 \\ \hline\end{array}$$

is declared fixed. Each such column contributes one plain i and one plain $i + 1$, so it is already balanced under the desired exchange. After this fixed segment, the algorithm moves upward and applies the same classification to the next untreated local block.

Attempt 2 (Formal involution argument for the eight cases)**.** *Each local rule was designed to have a prescribed inverse. The common $A_1 \cup B_1$ rule*

$$(i')^{\varepsilon} i^{a} (i+1)' (i+1)^{b} \longmapsto (i')^{\varepsilon} i^{b} (i+1)' (i+1)^{a}$$

is self-inverse. The count exchanges in A_4, B_2, B_4 and in the plain-i subcases of A_2, B_3 are also self-inverse. The remaining A_3 configurations are paired with the no-plain-i subcases of A_2 and B_3.

The proposed proof then proceeded inductively from bottom to top, keeping each following balanced vertical segment fixed. This was intended to give

$$\Phi_i^2(T) = T.$$

Example 3.3 (Why the bottom-up argument fails)**.** *The case-by-case calculation proves only that each prescribed local transformation has a formal inverse. It does not prove that the transformed filling is still an MSSYT: reordering a row can change the column inequalities, the left neighbour of a primed entry, and the skew-column depth of that entry. Thus preservation of (G1)–(G3) has not been established.*

More importantly, the second application need not use the same local decomposition. Reordering can move the first relevant occurrences of i and $i + 1$, thereby changing the Type A/Type B classification, the boundaries of the local blocks, and the fixed vertical segments. Consequently, the formal inverse of the original block may not be the rule applied on the second pass. Hence the local inverse checks do not establish a well-defined global involution on MSSYTs.

3.3 The DSVT Reformulation

The above Bender–Knuth proposal also suggestedrule induces a simpler transformation on DSVTs.

First, we divide the entries a and $a+1$ in a DSVT into two types. The first type consists of those entries for which a or $a+1$ is the smallest number in the corresponding box, namely, the entries that do not appear in the shadow part. The second type consists of those entries for which a or $a+1$ is not the smallest number in the corresponding box, namely, the entries that appear in the shadow part and are marked with a prime.

We fix all boxes in the DSVT that do not contain a or $a+1$. If there is a column containing two boxes, one containing a and the other containing $a+1$, we fix these two boxes. We also fix any box in a row that contains both a and $a+1$, provided that a is the smallest entry in the box and either there is no box to its right or the box immediately to its right contains an entry strictly smaller than a. We now consider only the boxes that have not been fixed.

The candidateabove Bender-Knuth involution induces the following transformation on the DSVT part was the following: in each row, we separately exchange the numbers of entries a and $a+1$ of the first type and the second type, and then reorder the entries according to the ordering condition of DSVT.

Already in a one-cell DSVT, the set $\{1,2\}$ has one minimum and one surplus entry. Swapping the numbers of minima and surplus entries independently can ask the same cell to have minimum In this way, the numbers of entries of each type remain unchanged. Clearly, once the numbers of $2a$ while retaining a surplus 1, which is impossible. Thus the two types are coupled by the set-valued cell structure; their row counts do not detand $a+1$ in the two types are determined, the resulting DSVT obtained by reordermine ag is unique valid output.

Example 3.4.

$$\begin{array}{|c|c|c|c|c|}\hline 3 & 23 & 2 & 2 & 1 \\ \hline 2 & 1 & 1 \\ \cline{1-3} 1 \\ \cline{1-1}\end{array} \quad \xrightarrow{\mathrm{Bk}_2} \quad \begin{array}{|c|c|c|c|c|}\hline 3 & 3 & 3 & 3 & 12 \\ \hline 2 & 1 & 1 \\ \cline{1-3} 1 \\ \cline{1-1}\end{array}$$

4 Main theorem

Theorem 4.1. *Let*

$$T_{\lambda,\mu} := \sum_{D \text{ Doubled}} \beta^{\mathrm{exc}(D)} \mathbf{x}^{\mathrm{content}(D)},$$

The sum over all doubled set-valued tableaux for the shapes (λ,μ) and box $k \times N$. Then

$$T_{\lambda,\mu} = G_\lambda \cdot G_\mu \mid_{\mathrm{box}} .$$

Proof. We construct a weight-preserving bijection between the doubled tableaux appearing in the definition of $T_{\lambda,\mu}$ and the compatible pairs appearing in

$$G_\lambda \cdot G_\mu \mid_{\mathrm{box}}$$

Given a doubled tableau D, assume T is the SVT in the upper left corner and U is the SVT in the lower right corner. We use the tableau $T * U$, obtained by rotating

U by 180° and then place it in the upper right corner of T. We then label all the boxes of $T * U$ starting from 1, reading from right to left and from top to bottom. First, we read all the elements of $T * U$ and arrange them in weakly increasing order to obtain I. We then read off the position labels of the corresponding elements in $T * U$ from I; among elements of the same value, their position labels are arranged in decreasing order. This gives the sequence A.

We apply Hecke insertion to (A, I) to obtain an increasing tableau P and a set-valued tableau Q. It suffices to show that Q fits inside a $k \times (n-k)$ rectangle if and only if T and U can be combined to form a $k \times (n-k)$ DSVT.

If T and U form a DSVT, let

$$A = (a_1, \dots, a_m), \qquad I = (i_1, \dots, i_m)$$

be the compatible pairs read from $T * U$, respectively. Suppose that the number i occurs k_i times in $T * U$, for $i = 1, \dots, k$. Then

$$i_1 = \cdots = i_{k_1}, \quad i_{k_1+1} = \cdots = i_{k_1+k_2}, \quad \cdots$$

and the compatibility conditions give

$$a_1 > a_2 > \cdots > a_{k_1}, \qquad i_{k_1+1} > \cdots > i_{k_1+k_2}, \quad \dots.$$

Each integer can occur at most $n-k$ times in $T * U$. Hence $k_i \le n-k$ for every i. The length of the longest decreasing subsequence of A equals the length of the first row of the increasing tableau P obtained from Hecke insertion. Since $\mathrm{shape}(P) = \mathrm{shape}(Q)$, it follows that the length of the first row of Q is also at most $n-k$. Hence Q fits inside a $k \times (n-k)$ rectangle.

Conversely, suppose that Q fits inside a $k \times (n-k)$ rectangle. Since $\mathrm{shape}(P) = \mathrm{shape}(Q)$, the first row of P has length at most $n-k$. By the correspondence between the longest decreasing subsequence of A and the first row of P, the longest decreasing subsequence of A has length at most $n-k$. In particular, for every i we have

$$k_i \le n-k.$$

Thus, each integer occurs at most $n-k$ times in $T * U$ hence the tableaux T and U can be combined to form a $k \times (n-k)$ DSVT.

□

5 MSSYTs and Genomic Tableaux

Genomic tableaux and MSSYTs both add extra information to a semistandard tableau. A genomic tableau divides equal entries into genes, while an MSSYT uses primes to record the nonminimal entries removed by column completion. The relation between these two descriptions is especially clear from their weights: an i-gene contributes one factor of x_i, and a plain i in an MSSYT also contributes one factor of x_i.

Definition 5.1 (Genomic tableau)**.** *Let S be an SSYT. For each i, divide the cells containing i into collections called* genes*. The division must satisfy the following conditions. If one gene contains entries in columns $c_1 < c_3$, then every i lying in* *a column strictly between c_1 and c_3 belongs to the same gene. Moreover, a gene contains at most one cell in each row.*

An SSYT together with such a division is called a genomic tableau*. Its weight is*

$$\mathbf{x}^{\mathrm{gwt}(S)} = \prod_{i\geq 1} x_i^{g_i(S)},$$

where $g_i(S)$ is the number of genes of family i.

The corresponding generating function is the genomic Schur function

$$U_\rho(\mathbf{x}) = \sum_{S\in\mathrm{Gen}(\rho)} \mathbf{x}^{\mathrm{gwt}(S)}.$$

The function U_ρ is symmetric, but it is not Schur-positive in general [3, Example 6.7]. By contrast, the marked Schur function is Schur-positive by the weight-complementing correspondence proved above. This is the sense in which MSSYTs provide a Schur-positive correction of the genomic-tableau model.

5.1 Coloring an MSSYT by genes

Let T be an MSSYT. Fix a value i and read all entries i and i' from west to east. Every plain i starts a new gene

$$i_1, i_2, \ldots,$$

and each i' receives the same label as the preceding plain i. If an i' occurs before every plain i, we label it i_0.

For example,

$$T = \begin{array}{|c|c|c|c|c|}\hline 1 & 1 & 2' & 2 & 4 \\ \hline 2 & 3' & 3 \\ \cline{1-3} 4' \\ \cline{1-1}\end{array} \qquad \longmapsto \qquad \widetilde{T} = \begin{array}{|c|c|c|c|c|}\hline 1_1 & 1_2 & 2_1 & 2_2 & 4_1 \\ \hline 2_1 & 3_0 & 3_1 \\ \cline{1-3} 4_0 \\ \cline{1-1}\end{array}.$$

The entries labelled 2_1 form one gene. The plain 2 starts this gene, and the entry $2'$ receives the same label. The entries $3'$ and $4'$ occur before any plain entry of the same value, so they receive the labels 3_0 and 4_0.

The labels i_j with $j \geq 1$ are ordinary gene labels. The label i_0 is an additional label used only for leading primed entries. In particular, $\widetilde{T}$ is a genomic tableau with possible virtual genes i_0.

The original MSSYT can be recovered from its coloring. In every gene i_j with $j \geq 1$, the first occurrence is decoded as a plain i, and the remaining occurrences are decoded as i'. Every occurrence of i_0 is decoded as i'. Thus the coloring gives a reversible encoding of the primes.

Moreover, the number of actual i-genes in $\widetilde{T}$ is exactly the number of plain entries i in T. Therefore

$$\mathbf{x}^{\mathrm{gwt}(\widetilde{T})} = \mathbf{x}^{\mathrm{content}(T)},$$

where the virtual genes i_0 are not included in $\mathrm{gwt}(\widetilde{T})$.

5.2 Genomic JDT on the colored tableau

Genomic jeu de taquin places bullets in the inner corners of a skew tableau and switches the bullets with one gene at a time. For a fixed gene G, all cells of G adjacent to bullets are switched simultaneously with those bullets. The genes are processed in their prescribed order.

We applied this operation to the colored form of an MSSYT:

$$T \longmapsto \widetilde{T} \xrightarrow{\text{genomic JDT}} \widetilde{R} \longmapsto R.$$

The first map is the coloring above, and the last map decodes the gene labels back into plain and primed entries.

Remark. *Why genomic JDT does not give an MSSYT JDT.* The coloring is reversible before any slides are performed, but genomic switches do not preserve the colored tableaux coming from MSSYTs. A switch may move an entry to a position where its decoded prime has a left neighbour of the same numerical value, violating (G2). It may also move a prime of value v to depth $d \geq v$, violating (G3).

We tried repairing such entries after rectification. This can produce a legal MSSYT, but legality alone is not enough. To obtain a K-theoretic Littlewood–Richardson rule, the tableaux rectifying to a fixed straight tableau must have the correct marked Schur generating function. The following example shows that this fails.

Example 5.2 (A rectification fiber with the wrong weight)**.** *Let*

$$\theta = (2,1,1)/(1)$$

and consider

$$T_1 = \begin{array}{|c|c|} \cline{2-2} \multicolumn{1}{c|}{} & 1 \\ \hline 2 & \multicolumn{1}{c}{} \\ \cline{1-1} 3 & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array}\,, \qquad T_2 = \begin{array}{|c|c|} \cline{2-2} \multicolumn{1}{c|}{} & 1 \\ \hline 2 & \multicolumn{1}{c}{} \\ \cline{1-1} 3' & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array}\,.$$

Under the final repair used in our genomic-JDT construction, both tableaux rectify to

$$R = \begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array}\,.$$

Their contributions are

$$\mathbf{x}^{\mathrm{content}(T_1)} = x_1x_2x_3$$

and

$$\beta^{\mathrm{exc}(T_2)}\mathbf{x}^{\mathrm{content}(T_2)} = \beta x_1x_2.$$

Hence this rectification fiber contains

$$x_1x_2x_3 + \beta x_1x_2.$$

For the alphabet $\{1,2,3\}$*, the only MSSYT of straight shape* $(1,1,1)$ *is the plain column* R*. Indeed, marking the entry* v *in this column would place* v' *at depth* v*, contradicting (G3). Therefore*

$$\mathfrak{g}_{(1,1,1)}(x_1,x_2,x_3;\beta) = x_1x_2x_3.$$

The extra term $\beta x_1 x_2$ shows that the rectification fiber does not have the required marked Schur generating function. Thus the genomic switches, even with the terminal repair, do not define the desired MSSYT rectification.

6 Toric Schur Polynomials

The preceding sections attempted to define Bender–Knuth and jeu de taquin operations directly on MSSYTs. In both constructions, the main obstruction came from the primed entries: moving a prime may destroy its depth condition or separate it from the plain entry that supports it. We therefore consider a different direction which keeps the original set-valued data of a DSVT and places it in a periodic diagram.

6.1 From a planar shape to a toric shape

Postnikov [4] considers diagrams on the cylinder

$$\mathcal{C}_{k,n} = \mathbb{Z}^2/(-k, n-k)\mathbb{Z}.$$

Thus a lattice path is extended periodically under the shift

$$(i, j) \longmapsto (i-k, j+n-k).$$

A cylindric shape $\lambda/d/\mu$ is determined by two periodic boundary paths. Starting with the paths $\lambda[0]$ and $\mu[0]$ in a $k \times (n-k)$ rectangle, shift $\lambda[0]$ by d steps and extend both paths periodically. The cells between the shifted path $\lambda[d]$ and the fixed path $\mu[0]$ form the cylindric shape $\lambda/d/\mu$.

The construction is shown schematically in Figure 2. The middle diagram is the essential step: the two paths are treated as periodic paths, so a region leaving one side of the rectangle continues from the opposite side.

$$k = 4, \qquad n = 10, \qquad \lambda[d] = (3,2,1,1)[2], \qquad \mu = (4,4,2,1)$$

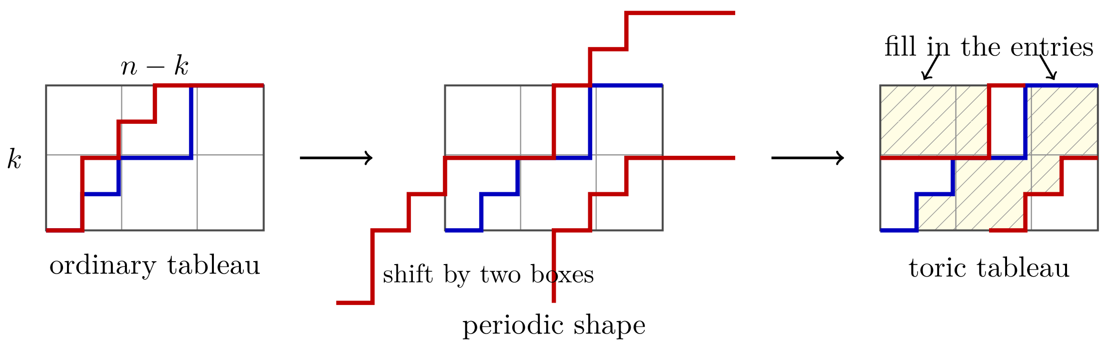

Figure 2: The passage from the planar boundary paths to the periodic shape. The red path is shifted by two boxes and extended periodically, while the blue path remains fixed. The hatched cells in the final fundamental rectangle are precisely the cells between the two periodic paths.

A semistandard toric tableau is weakly increasing along each periodic row and strictly increasing down each periodic column. The comparisons are read in the periodic diagram, so a row or column may continue across the boundary of the displayed rectangle.

Postnikov defines the toric Schur polynomial by

$$s_{\lambda/d/\mu}(x_1, \ldots, x_k) = \sum_S \mathbf{x}^{\mathrm{content}(S)},$$

where S ranges over all semistandard toric tableaux of shape $\lambda/d/\mu$. When $d = 0$, the shape is the ordinary skew shape λ/μ, and therefore

$$s_{\lambda/0/\mu} = s_{\lambda/\mu}.$$

6.2 Connection with DSVTs

The MSSYTs appearing in this report have shape

$$\theta = \mu^\vee/\lambda = \mu^\vee/0/\lambda.$$

Thus the shape used in our construction is already the $d = 0$ case of the toric family

$$\mu^\vee/d/\lambda.$$

A DSVT retains the set-valued entries that are recorded by primes in the corresponding MSSYT. This suggests extending Postnikov's construction to toric set-valued tableaux. For a toric shape κ, each cell a would contain a nonempty set $\mathcal{T}(a) \subseteq [k]$, with

$$\max \mathcal{T}(a) \le \min \mathcal{T}(b)$$

along rows and

$$\max \mathcal{T}(a) < \min \mathcal{T}(c)$$

down columns. These inequalities would also be imposed when the adjacent cells lie on opposite sides of the displayed rectangle.

The resulting generating function would be

$$G_\kappa^{\mathrm{tor}}(\mathbf{x}; \beta) = \sum_{\mathcal{T}} \beta^{\mathrm{exc}(\mathcal{T})} \mathbf{x}^{\mathrm{content}(\mathcal{T})},$$

where

$$\mathrm{exc}(\mathcal{T}) = \sum_{a \in \kappa} \big(|\mathcal{T}(a)| - 1\big).$$

At $\beta = 0$, only singleton-valued cells remain, and this generating function reduces to Postnikov's toric Schur polynomial.

Postnikov proved the expansion

$$s_{\mu^\vee/d/\lambda}(x_1, \ldots, x_k) = \sum_{\nu \in P_{k,n}} C_{\lambda\mu}^{\nu,d} s_{\nu^\vee}(x_1, \ldots, x_k).$$

For $d = 0$, the coefficients are the ordinary Littlewood–Richardson coefficients. This shows that the single-valued specialization of our shape already belongs to the toric Schur framework.

The main theorem proved above through compatible pairs and Hecke insertion is

$$T_{\lambda,\mu}(\mathbf{x};\beta) = G_\lambda(\mathbf{x};\beta)G_\mu(\mathbf{x};\beta)\big|_{\text{box}}.$$

The proposed toric model asks whether this generating function also has an interpretation of the form

$$T_{\lambda,\mu}(\mathbf{x};\beta) \stackrel{?}{=} G^{\text{tor}}_{\mu^\vee/d/\lambda}(\mathbf{x};\beta).$$

To construct such a correspondence, one must determine the displacement d associated with a DSVT and verify that the DSVT conditions become the semistandard inequalities on the periodic shape. This would keep the nonminimal entries inside their original set-valued cells and avoid the local movement of primes that caused difficulties in the Bender–Knuth and jeu de taquin constructions.

References

[1] A. S. Buch, *A Littlewood–Richardson rule for the K-theory of Grassmannians*, Acta Math. **189** (2002), 37–78.

[2] H. Thomas and A. Yong, *Longest increasing subsequences, Plancherel-type measure and the Hecke insertion algorithm*, Adv. in Appl. Math. **46** (2011), 610–642.

[3] O. Pechenik and A. Yong, *Genomic tableaux*, J. Algebraic Combin. **45** (2017), no. 3, 649–685.

[4] A. Postnikov, *Affine approach to quantum Schubert calculus*, Duke Math. J. **128** (2005), no. 3, 473–509.

[5] E. Marberg, K. H. Tong, and T. Yu, *Grothendieck positivity for normal square root crystals*, Adv. Math. **501** (2026), Article 111118.

[6] S. Kundu, *A contratableau model for the K-theoretic Littlewood–Richardson rule*, Electron. J. Combin. **33** (2026), no. 3; arXiv:2504.02290.